

Over the past decade, the technology sector, represented by the NASDAQ-100 ETF (QQQ), has played a pivotal role in driving U.S. equity market performance. Our project set out to analyze how QQQ has performed relative to the broader S&P 500 ETF (SPY) from 2015 through 2025, with the goal of quantifying both the magnitude and nature of this outperformance, as well as the associated risks. The motivation for this analysis stems from the increasing prominence of technology stocks in investor portfolios and the need to understand the trade-offs involved in sector-specific investing.

To conduct this study, we gathered historical adjusted closing prices for both QQQ and SPY, calculating cumulative returns to directly compare long-term growth. We then computed rolling one-year compounded returns to capture the evolution of annual performance over time, and rolling one-year annualized volatility to assess changes in risk. Finally, we estimated QQQ's Beta relative to SPY, arriving at a value of 1.1534, which quantifies QQQ's sensitivity to broad market movements. These metrics were visualized in a series of charts to facilitate direct comparison and highlight key periods of divergence and convergence between the two ETFs.

The results of our analysis are striking. QQQ's cumulative returns far outpaced those of SPY, particularly during the period from 2019 to 2021, when technology stocks benefited from strong earnings growth and pandemic-driven demand for digital services. At its peak, QQQ achieved over four times its starting value, while SPY's cumulative return was closer to two and a half times, as shown in our first chart. However, this outperformance came with greater volatility: QQQ's rolling annualized volatility consistently exceeded that of SPY, with pronounced spikes during periods of market stress, most notably the 2020 pandemic selloff and the 2022–2023 correction. The rolling one-year return chart further illustrates that while QQQ's upside was greater during bull markets, its drawdowns were also more severe during downturns, underscoring the higher risk profile of the technology sector.

The calculated Beta of 1.1534 reinforces this conclusion, indicating that QQQ is more responsive to market movements than SPY and amplifies both gains and losses. Notably, the period from 2020 to 2021 saw QQQ's annual returns and volatility surge to historic highs, reflecting the sector's exposure to both the benefits and risks of rapid innovation and changing economic conditions. In more recent years, the gap between QQQ and SPY narrowed somewhat, as technology stocks faced headwinds from rising interest rates and shifting market sentiment.

While our findings highlight the long-term benefits of allocating to technology, they also reveal important limitations. Our analysis is based solely on historical price data, and while it uses adjusted closing prices, it may not fully capture the impact of dividends or changes in index composition. Additionally, the study does not address survivorship bias or macroeconomic factors that could explain shifts in volatility and returns. Future research could extend this work by incorporating other sector ETFs, analyzing the impact of economic variables, or applying multi-factor risk models for deeper attribution.

In summary, the technology sector delivered superior returns relative to the broader market over the past decade, but with meaningfully higher volatility and risk. Investors seeking to overweight technology must therefore balance the potential for outsized gains against the likelihood of larger drawdowns, particularly during periods of market turbulence.